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[GS PAPER II – POLICIES AND REGULATORY FRAMEWORK • GS PAPER III – ENVIRONMENT AND CLIMATE CHANGE, ENERGY INFRASTRUCTURE, ECONOMY AND ELECTRICITY MARKETS]

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Solar Surge, Grid Squeeze



Understand how India's rapid solar and renewable energy expansion is straining transmission infrastructure and energy storage systems. Explore solutions such as grid modernisation, battery storage and electricity market reforms to ensure reliable, affordable clean power.

Solar Surge, Grid Squeeze

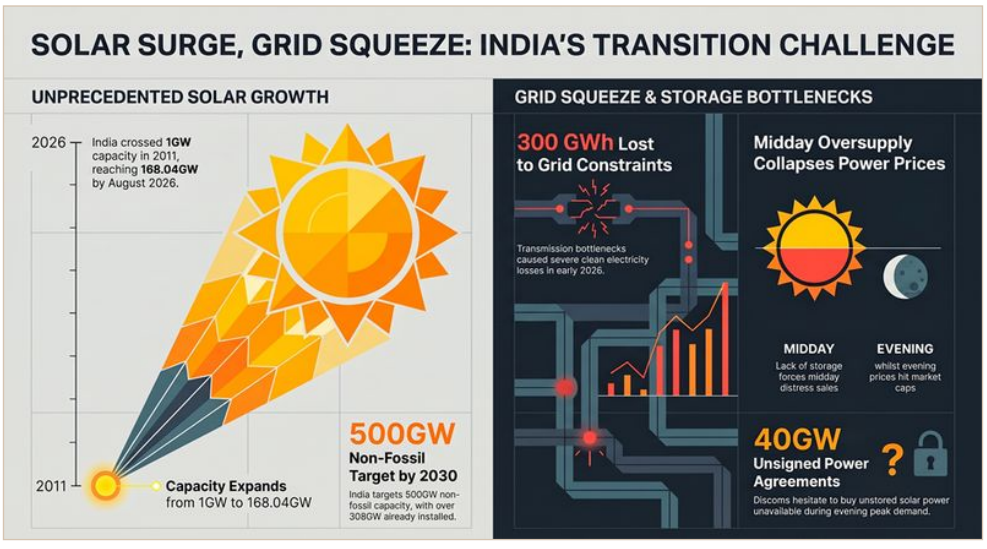
UPSC linkage:

GS Paper II – Policies and Regulatory Framework

GS Paper III – Environment and Climate Change, Energy Infrastructure, Economy and Electricity Markets

Strengthening Electricity Infrastructure

As India's green energy drive enters its next phase, the focus has turned to strengthening the country's electricity infrastructure. This includes building adequate capacity to carry electricity from solar and wind farms to the power grid and then to factories, homes and other consumers.



India's Solar Expansion

For a country that recorded all of the world's 50 hottest cities, India's ability to harness the sun's energy took decades to take off.

It wasn't until 2011-12 that the country crossed a critical milestone: 1 gigawatt (GW) of installed solar power capacity. The next significant milestone was 100GW, achieved more than a decade later, in early 2025.

India has galloped since, adding nearly 50GW of solar power capacity in 2025 and reaching 168.04GW by 31 August 2026.

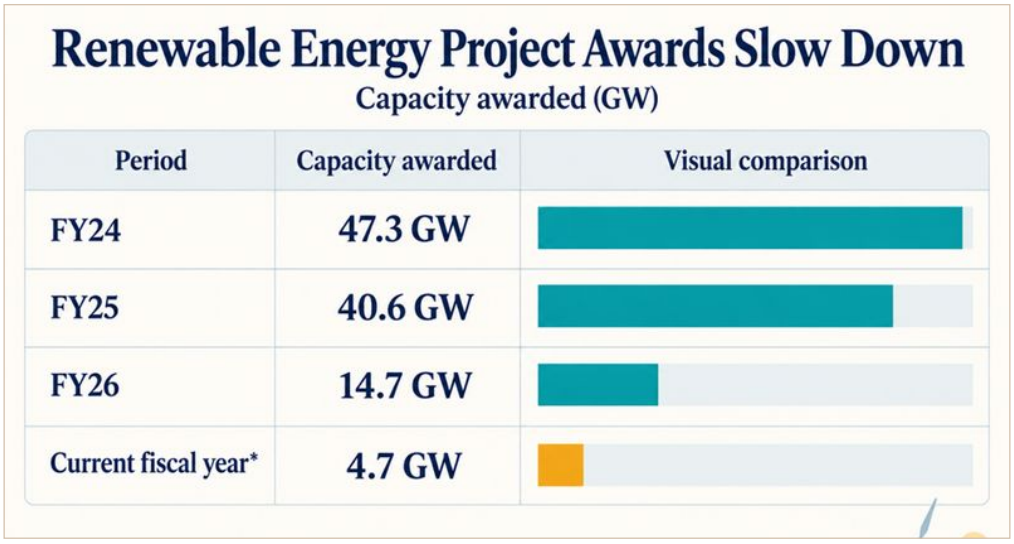
Transmission and Storage Challenges

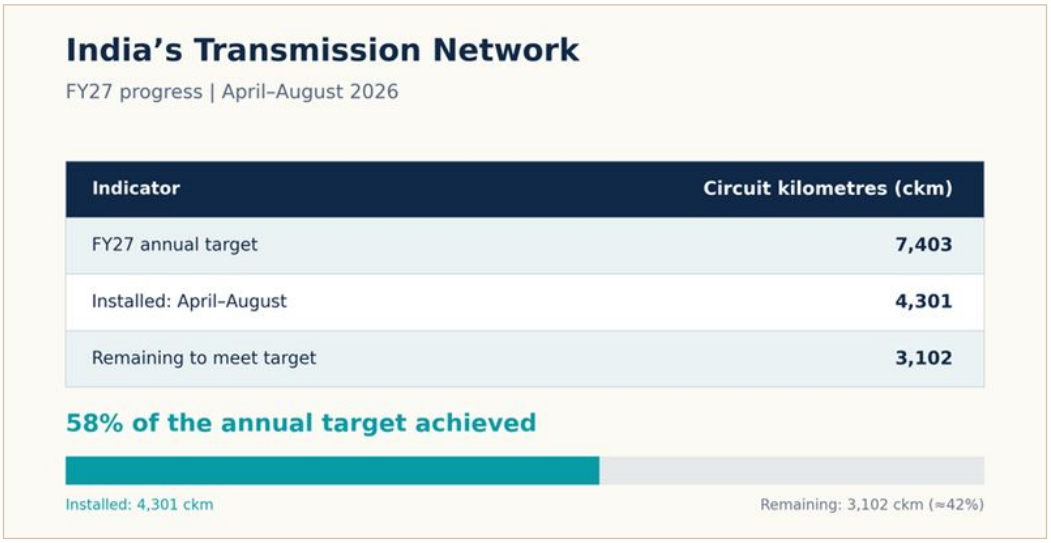
India's transmission infrastructure is struggling to carry all the electricity generated when the sun shines. Surplus solar power can congest the grid, while inadequate storage means excess energy cannot always be saved for use when demand rises.

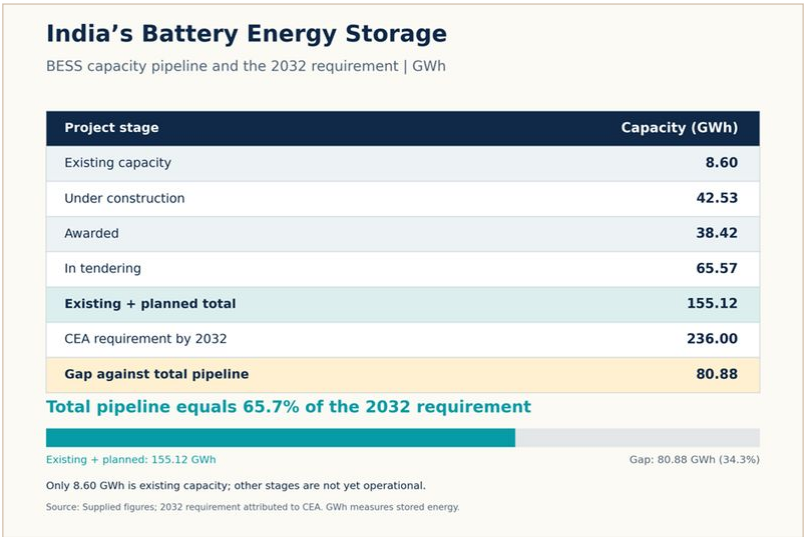
At the same time, fluctuations in solar power generation, depending on the time of day and weather, make it harder to balance electricity supply and demand, and maintain the grid's frequency within the permitted range.

One of the world's largest energy transition efforts—from fossil fuels to clean energy—by the world's third-largest generator of renewable energy.

CHANGING PRIORITIES







India's Battery Energy Storage

BESS capacity pipeline and the 2032 requirement | GWh

Project stage	Capacity (GWh)
Existing capacity	8.60
Under construction	42.53
Awarded	38.42
In tendering	65.57
Existing + planned total	**155.12**
CEA requirement by 2032	236.00
Gap against total pipeline	**80.88**

Total pipeline equals 65.7% of the 2032 requirement

Existing + planned: 155.12 GWh

Gap: 80.88 GWh (34.3%)

Only 8.60 GWh is existing capacity; other stages are not yet operational.

Source: Supplied figures; 2032 requirement attributed to CEA. GWh measures stored energy.

Pumped Storage and Surplus Power

India also has a pipeline of 122.77GW of pumped storage capacity at various stages of development. These plants can absorb surplus renewable power during off-peak hours and release it when demand rises. The CEA has a roadmap to install another 100GW of pumped storage project (PSP) capacity by 2035-36.

Inadequate energy storage has led to an oversupply of solar power, forcing power generators and discoms into distress sales.

SUSTAINING MOMENTUM

Renewable Capacity Addition

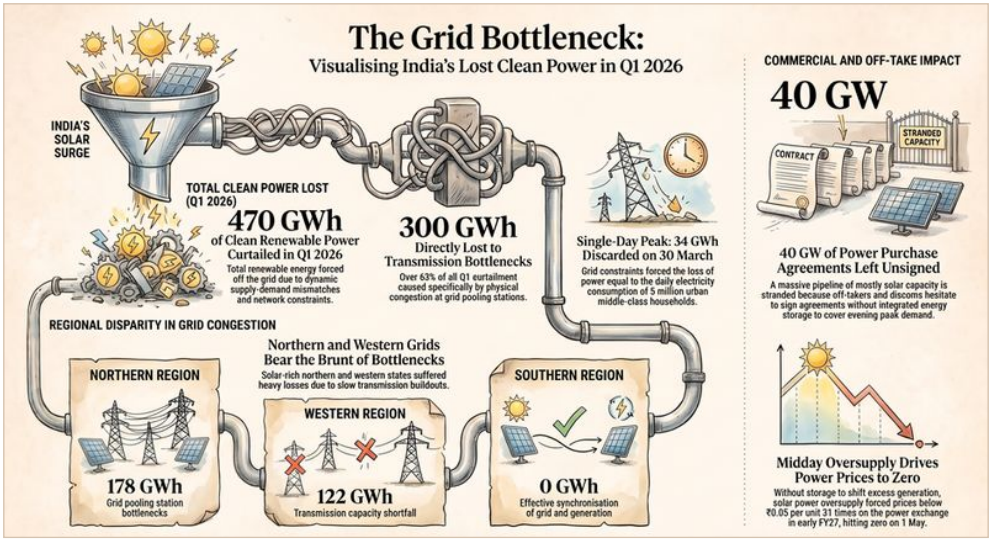
In the first six months of FY27 (April–September), India added 24.56GW of renewable energy capacity, including 20.79GW of solar and 3.11GW of wind capacity, show data from the ministry of new and renewable energy (MNRE).

Overall, India had 308.02GW of installed non-fossil fuel-based electricity capacity as of 30 September, including 299.24GW of renewable energy capacity. Another 183.14GW is under implementation, while 51.29GW has been tendered. India is targeting 500GW of non-fossil fuel-based power capacity by 2030.

Grid Integration and Coordination

The ministry's focus was on expanding transmission infrastructure, improving forecasting and scheduling, and coordinating with the ministry of power, the CEA, state governments and utilities to minimise congestion and renewable energy curtailment.

India's energy transition is moving from capacity addition towards delivering reliable, affordable around-the-clock green power.



Storage-backed Renewable Power

We are promoting round-the-clock (RTC), firm and dispatchable renewable energy (FDRE), and solar-wind hybrid projects integrated with storage to ensure that renewable electricity is available in line with demand, including during evening and peak hours. Battery energy storage systems and pumped storage hydropower will play a critical role in balancing the variability of renewable energy.

Transmission Planning for the Next Phase

The CEA's March 2026 transmission plan envisages transmission infrastructure to integrate more than 900GW of non-fossil capacity by 2035-36.

India's energy transition is entering a more demanding phase. The focus is no longer limited to adding renewable capacity; it is equally about ensuring the grid can integrate and deliver that clean power when and where it is needed.

COST OF CONSTRAINTS

Clean Electricity Lost to Congestion

According to Ember, a Norway-based think tank focused on energy transition, transmission constraints led to the loss of 300 GWh of clean electricity in India in the first quarter of 2026 (January–March), particularly at grid pooling stations in the northern and western regions.

On 30 March alone, India lost 34 GWh of clean generation, equivalent to the daily electricity use of about 5 million urban middle-class households.

Regional Patterns of Curtailment

The analysis also found that total renewable energy curtailment in Q1 2026 reached around 470 GWh, of which 300 GWh stemmed directly from transmission constraints. The northern region accounted for 178 GWh and the western region for 122 GWh. The southern region recorded none, reflecting better synchronisation between generation and grid expansion.

Unsigned Power Agreements

Inadequate transmission capacity has left about 40GW of power purchase agreements (PPAs) and power sale agreements (PSAs) unsigned, according to the Ember report. This 40GW of capacity, mostly solar, lacks integrated storage and is therefore less attractive to electricity distribution companies, or discoms, as the power is unavailable in the evening when demand is usually higher.

ZERO PRICES

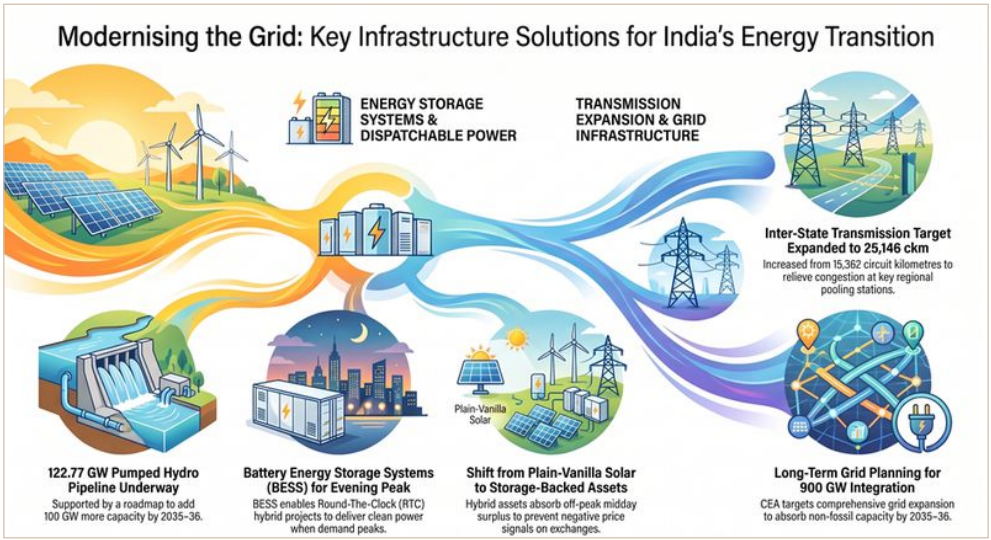
Transmission Targets and Connectivity Delays

India has met only about 80% of its annual transmission targets over the past five years, according to Ember. As a result, the target for the inter-state transmission system, which is crucial for evacuating power from solar-rich states to other states, has increased from 15,382 circuit kilometres to 25,146 ckm for FY27.

Ember added that about 20GW of renewable energy capacity is likely to face connectivity delays of more than four months in FY27 because of slow installation of transmission capacity.

Midday Oversupply and Distress Sales

Inadequate energy storage has also led to an oversupply of solar power, primarily between 11 am and 3.30 pm, forcing discoms into distress sales on power exchanges, according to executives with power developers.



Power prices fell below Rs. 0.05 per unit 31 times in the first quarter of FY27, hitting zero on 1 May, according to data from the Indian Energy Exchange. In the year-earlier first quarter, power prices had dropped below Rs. 0.05 per unit six times.

The Market Signal for Storage

Zero prices on the exchange are “a symptom of generation success running ahead of storage infrastructure”, added Sanjeev Aggarwal, founder and chairman, Hexa Climate Solutions, a renewable power generator. “It is a loud, clear market signal that we urgently need to shift this excess energy to the evening peak. It isn’t a demand problem; it’s a grid-balancing and storage problem.”

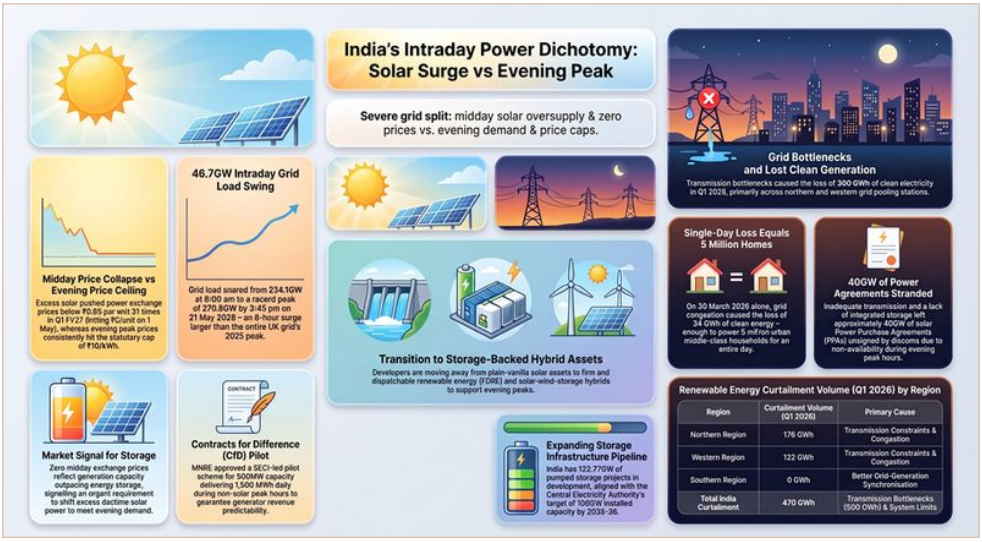
POWER DICHOTOMY

Rising Demand, Falling Prices

The decline in power prices comes despite rising electricity demand. India’s peak power demand hit a record 270.8GW on 21 May and is expected to reach 272GW this fiscal year. The power ministry projects peak demand to rise further to 300GW in FY28.

Intraday Demand Volatility

Noting this dichotomy, Sanjeev Sanyal, a member of the Economic Advisory Council to the Prime Minister (EAC-PM), said in a recent working paper that India’s power demand is not only large but also extremely volatile even within the duration of a single day.



"On 21 May 2026, India's power grid set an all-time record. At 3:45 pm, the peak demand met reached 270.8GW. As striking as the quantum of the load was the speed of its change. At 8 am on the same day, demand sat at 224.1GW. In less than eight hours, the grid took on an additional load of 46.7GW, more than the entire British grid drew at its peak in all of 2025," stated the working paper titled 'The Duck and The Camel: Tracing the Net Load on the Indian Power Grid', co-authored by Sanyal and Satvik Dev, joint director, EAC-PM.

The Gap Between Demand Peaks and Price Peaks

Peak demand and peak power prices often occur at different times of the day, and the gap between them has become an important feature of the modern Indian grid, the authors stated. And the best way to smooth these fluctuations, they added, is to store electricity when it is abundant and release it when it is scarce.

Sanyal and Dev noted that power prices fell to near zero per unit during peak solar hours but hit the price cap of Rs. 10 per kWh in the evening, when solar power is unavailable.

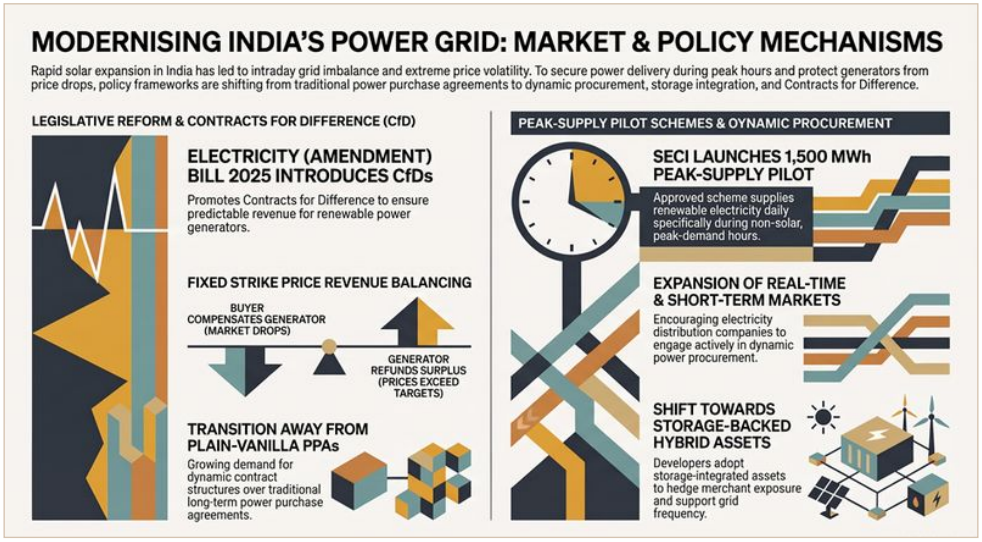
Contracts for Difference: Revenue Predictability

To ensure predictable revenue for power generators, the Electricity (Amendment) Bill, 2025 proposes promoting 'contracts for difference' (CfD).

Under this arrangement, a power generator and a buyer agree on a fixed reference price, known as the strike price. If the market price falls below that level, the buyer pays the difference to the generator. If it rises above the strike price, the generator refunds the difference.

Rewarding Flexibility

Sanyal and Dev made a case for incentivising flexibility, noting that energy storage, demand response and the ability to ramp up power generation quickly are precisely the resources whose value lies in intraday price spreads. A market dominated by long-term energy contracts gives power generators little opportunity to monetise that value.



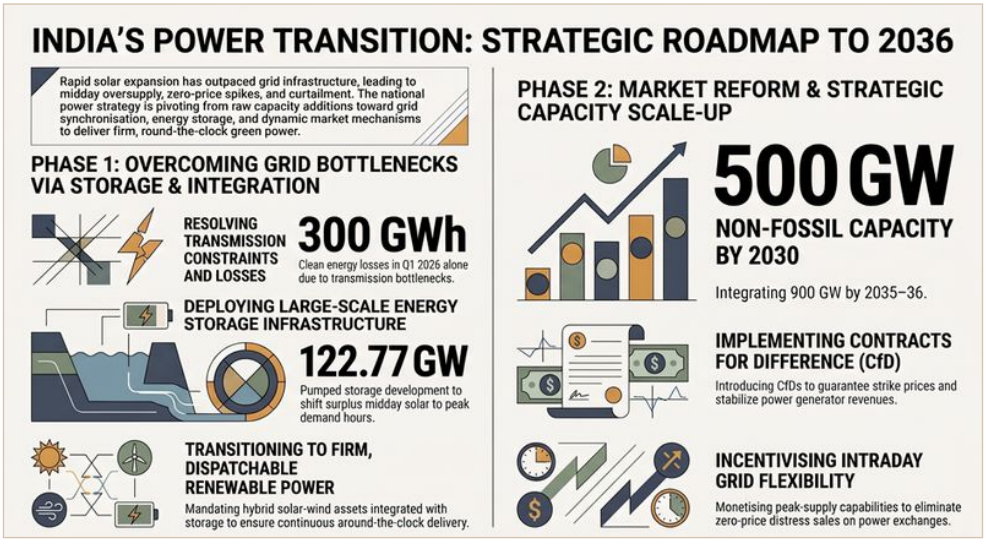
Regulatory Recognition for Storage

Empowering the regulator to design market products is one of the institutional preconditions for expanding storage capacity, they said, insisting that the most direct response to the flexibility problem is to give legal standing to storage systems.

Currently, most power market transactions are based on power purchase agreements, but sector experts say demand is growing for new procurement models, CfD being a key alternative.

SECI's Pilot CfD Scheme

In a notification issued in March, MNRE approved a pilot CfD scheme, designating the Solar Energy Corp. of India (Seci) as the nodal agency for 500 megawatt (MW) of capacity supplying 1,500 MWh of renewable electricity daily during non-solar hours.



Towards Dynamic Power Procurement

"India now needs to deepen short-term and real-time markets, bring discoms more actively into dynamic procurement, and scale policy moves such as Seci's 1,500 MWh CfD-based peak-supply tender much faster," said Shashank Sharma, founder and CEO, Sunsure Energy, a commercial and industrial-focused renewable energy developer.

"For developers, the message is equally clear: the old model of plain-vanilla solar spilling excess power into the exchange is giving way to hybrid, storage-backed assets that shift clean power into evening peaks, support the grid and hedge merchant exposure"

Mains question:

GS Paper III | 15 Marks

“India’s renewable energy transition faces challenges of grid integration and energy storage.” Discuss.

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